

Final Exam

ISE 5970: Advanced Systems Optimization

School of Industrial and Systems Engineering

University of Oklahoma

Spring 2025

Instructions for Submission:

1. The solution report must be uploaded to Canvas by the indicated due date. The solution should be uploaded to Canvas as a PDF file.
2. Along with your main submission (in PDF format), please ensure you also attach any supplementary files (Excel, Python) that were instrumental in your work. If any Python code was used, upload it in HTML format.

Additional details

- Total points: 100 + 30 points extra credit
- You may use your notes, textbooks, and slides. No internet resources or communication with others. You may use Gurobi/Python/Excel.
- Show all your work. Justifications and interpretations are as important as correct calculations.

Problem Context

A logistics company is deciding whether to open or not some potential distribution centers (I) to meet the demands of a list of different products (K) for a group of customer regions (J). Assume that there is a fixed cost f_i of opening distribution center $i \in I$, and a cost c_{ijk} of supplying the full demand of client $j \in J$ for product $k \in K$ with distribution center $i \in I$, as given in Table 1

Distribution Center	Customer region					Fixed cost
	1	2	3	4	5	
1	2, 4	3, 1	4, 2	5, 5	7, 2	2
2	4, 3	3, 1	1, 1	2, 5	6, 3	3
3	5, 2	4, 5	2, 2	1, 3	3, 3	3

Table 1: Fixed cost f_i , and a cost c_{ijk} . for each pair i, j , commas separate the supplying cost between different types of product k .

Exam Questions (100 points total + 30 points extra credit)

(a) (15 points) Modeling and Understanding

Clearly write a general optimization problem for this problem, including sets, parameters, decision variables, objective function, and constraints. Briefly explain the role of each type of variable and constraint. For the model, use the same structure and notation from the

allocation problem discussed in class to illustrate Benders decomposition, but now extending it to a multicommodity problem.

(b) (15 points) **Dual problem**

Write the dual problem associated with the previous model, if integrality of the variable for opening each distribution center was relaxed. Discuss the meaning of each constraint, variable, and objective function, in the context of the problem.

(c) (10 points) **Method Selection**

During the semester we discussed diverse methods to solve optimization models, including:

- Benders Decomposition
- Dantzig-Wolfe Decomposition
- Lagrangian Relaxation
- Branch and Bound
- Gomory Cuts / Cutting Planes

For each of them, discuss whether it would be applicable or not for this problem. Which of the following decomposition or relaxation methods would be most suitable for this problem? Could you combine them?

(d) (10 points) **Scalability and Extensions**

Briefly discuss how the method(s) that apply would scale, if:

- The number of facilities increases
- The number of customers increases
- The number of products increases

What are the real-world implications?

(e) (25 points) **Benders Decomposition:**

Solve the original problem (part a) using Benders decomposition. Show the detailed results of each iteration, and plot the upper and lower bounds as a function of the iteration, until they converge. Start your iterations by fixing $y = (1, 0, 0)$. For each iteration, indicate the type of Benders cut added.

(f) (25 points) **Dantzig-Wolfe Decomposition:**

Write the Dantzig-Wolfe Decomposition/Column generation algorithm for the subproblem used in the Benders Decomposition (part d). Fully solve the first Benders iteration subproblem (i.e., where you fixed $y = (1, 0, 0)$) using Dantzig-Wolfe, showing each step in detail.

(g) (10 points extra) **Lagrangian relaxation:**

Perform a Lagrangian relaxation to the original problem, relaxing the demand satisfaction constraints. Show all the steps made, starting from indicating the new problem to solve, and the results for each iteration. Make plots of (i) how each Lagrangian multiplier vs each iteration, as well as (ii) the objective function vs each iteration.

(h) (10 points extra) **Branch-and-Bound:**

Solve the original problem using Branch-and-Bound. Show your tree in detail.

(i) (10 points extra) **Cutting planes:**

Solve the original problem using cutting planes method. Show your iterations in detail.